Double angle identities

1 Evaluate:

a $\cos^2(15^\circ) - \sin^2(15^\circ)$

b $2\sin 22.5^\circ \cos 22.5^\circ$

c $2\cos^2(157.5^\circ) - 1$

d $1 - 2\sin^2(15^\circ)$

e $10\sin 67.5^\circ \cos 67.5^\circ$

f $\frac{1}{6}\sin 157.5^\circ \cos 157.5^\circ$

g $3\cos^2(165^\circ) - 3\sin^2(165^\circ)$

h $\frac{1}{4} - \frac{1}{2}\sin^2(67.5^\circ)$

2 Use the double angle identity for the cosine ratio to simplify the following expression:

$$10\cos^2\left(\frac{5\pi}{8}\right) - 5$$

3 Given $\cos\theta = \frac{4}{5}$ and $\sin\theta < 0$, find:

a $\sin\theta$

b $\sin 2\theta$

c $\cos 2\theta$

4 Given $\sin\theta = -\frac{7}{25}$ and $\cos\theta < 0$, find:

a $\cos\theta$

b $\sin 2\theta$

c $\cos 2\theta$

5 Given $\cos\theta = \frac{4}{5}$ and $\sin\theta = \frac{3}{5}$, find:

a $\sin 2\theta$

b $\cos 2\theta$

6 Consider the expression $\frac{1+\cos x}{2}$ where $\sin x = -\frac{12}{13}$ and $\frac{3\pi}{2} < x < 2\pi$.

a Simplify $\frac{1+\cos x}{2}$.

b Find $\cos x$.

c Hence find $\cos\left(\frac{x}{2}\right)$.

7 Consider the expression $\frac{1+\cos x}{2}$ where $\sin x = \frac{5}{13}$ and $\frac{\pi}{2} < x < \pi$.

a Simplify $\frac{1+\cos x}{2}$.

b Find $\cos x$.

c Hence find $\cos\left(\frac{x}{2}\right)$.

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8 Consider the expression $\frac{1-\cos x}{2}$ where $\tan x = 4$ and $0 < x < \frac{\pi}{2}$.

a Simplify $\frac{1-\cos x}{2}$.

b Find $\cos x$.

c Hence find $\sin\left(\frac{x}{2}\right)$.

9 Find the exact value of the following:

a $\sin\frac{3\pi}{8}$ **b** $\cos\frac{7\pi}{12}$ **c** $\sin 67.5^\circ$ **d** $\cos 157.5^\circ$

10 Find the exact value of the following with the given conditions:

a $\sin\left(\frac{x}{2}\right)$ if $\cos x = -\frac{4}{7}$ and $\frac{\pi}{2} < x < \pi$

b $\cos\left(\frac{x}{2}\right)$ if $\cos x = \frac{2}{5}$ and $0 < x < \frac{\pi}{2}$

c $\cos x$ if $\cos 2x = \frac{2}{3}$ and $\pi < x < \frac{3\pi}{2}$

d $\sin x$ if $\cos 2x = -\frac{4}{7}$ and $\frac{3\pi}{2} < x < 2\pi$

11 If $\tan x = \frac{p}{q}$, express $p\sin 2x + q\cos 2x$ in terms of p and q only.

12 Consider the expression $\cos 3\theta$.

a By writing $\cos 3\theta$ as $\cos(2\theta + \theta)$, show that $\cos 3\theta = 4\cos^3(\theta) - 3\cos\theta$.

b Using the fact that $\sin 36^\circ = \cos 54^\circ$, solve for the exact value of $\sin 18^\circ$.

13 Prove the following by simplifying the left hand side (LHS) of the identity:

a $\frac{1-\cos 2x}{\sin 2x} = \tan x$

b $\frac{2\sin^3(x) + 2\cos^3(x)}{\sin x + \cos x} = 2 - \sin 2x$

c $\sin^4(x) + \cos^4(x) = 1 - \frac{1}{2}\sin^2(2x)$

d $\frac{\sin 2x + 1}{\cos 2x} = \frac{\cos x + \sin x}{\cos x - \sin x}$

e $\frac{\sin^3(A) + \cos^3(A)}{\sin A + \cos A} = 1 - \frac{1}{2}\sin 2A$

f $\frac{2\cos\left(\frac{x}{2}\right) - 1 - \cos x}{2\cos\left(\frac{x}{2}\right) + 1 + \cos x} = \frac{1-\cos\left(\frac{x}{2}\right)}{1+\cos\left(\frac{x}{2}\right)}$

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Equations

14 Solve the following equations. Round your answer to one decimal place when necessary.

a $\cos 2x = 3\cos x - 2$ for $0 \leq x < 2\pi$

b $\sin 2x - \cos x = 0$ for $0 \leq x \leq 2\pi$

c $3\sin x - 1 = 1 - \cos 2x$ for $0 \leq x \leq 2\pi$

d $\cos^2(x) = 2\cos^2\left(\frac{x}{2}\right)$ for $0^\circ \leq x \leq 360^\circ$

e $8\sin^2(x) - 4\sin 2x = 8$ for $0^\circ \leq x \leq 360^\circ$

15 Consider the identity $\sin(\theta + 2\theta) = \sin\theta\cos 2\theta + \cos\theta\sin 2\theta$.

a Prove that $\sin 3\theta = 3\sin\theta - 4\sin^3(\theta)$.

b Hence solve the equation $\sin 3\theta = 2\sin\theta$ for $0 \leq \theta \leq \pi$.

16 Consider the equation $x^2 - \sqrt{2}\sin(45^\circ + \theta)\,x + \sin\theta\cos\theta = 0$.

a Find the value of the discriminant of the equation.

b What can be said about the equation's solutions?

Applications

17 The speed of an aircraft can be measured by a 'Mach number', m. The Mach number is the ratio of the speed of an aircraft to the speed of sound. For example, a speed of Mach 2 means that the aircraft is travelling at 2 times the speed of sound.
If an aircraft is travelling faster than the speed of sound ($m > 1$), it projects sound waves that form a cone behind it. If the angle at the vertex of the cone is θ, then

$$\sin\left(\frac{\theta}{2}\right) = \frac{1}{m}$$

a If $m = \frac{3}{2}$, then find the value of θ to the nearest degree.

b If $\theta = 90$, then find the value of m to one decimal place.

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18 The power, P, consumed by an electrical device can be calculated by taking the product of the current, I, and voltage, V, in its circuit. That is, $P = VI$.

When a light is turned on, a household circuit has current $I = \frac{5}{4}\sin 110\pi t$ and voltage $V = 151\sin 110\pi t$ at time t.

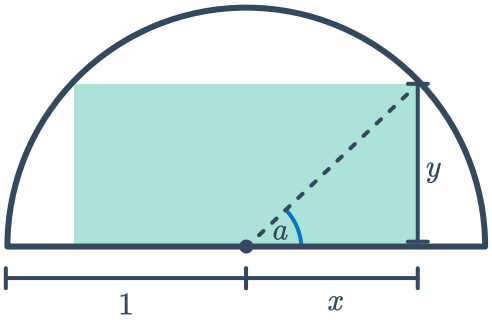
a Find the equation for the power consumed by the light, P, in terms of t.

b Use your CAS calculator to graph the function $P(t)$, and find the maximum and minimum values for power consumed by the light.

c Using a double angle rule, express P in the form $P = a\cos b\pi\theta + c$.

d The power rating of a lightbulb is the average of the maximum and minimum amount of power that is used by the lightbulb, rounded to the nearest 10 watts. Find the power rating of the lightbulb.

19 A rectangle is inscribed in a semicircle of radius 1, as shown in the diagram:



a Express x in terms of a.

b Express y in terms of a.

c Let A be the area of the triangle. Show that $A = \sin 2a$.

d Notice that a must be between 0° and 90°. What value of a makes the area of the rectangle largest?

e Hence, find the value for y which results in the largest possible area. Give your answer as an exact value with a rational denominator.

f Hence, find the length of the rectangle which results in the largest possible area. Give your answer as an exact value with a rational denominator.

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